

Linus Torvalds. Although patches for x86-64 support have been available for a while, the move by SuSE would provide an official stamp by writing support directly into the kernel. Linux support is extremely crucial for SledgeHammer, more than for ClawHammer, if AMD wishes to gain traction with corporations that purchase heaps of server technology. In fact, this has been the goal of AMD for years. Taking advantage of Linux—an OS that is already popular in the server circles—could be regarded as a smart move by AMD in this direction. Note that Linux already works on several 64-bit chips such as the Compaq Alpha and Sun UltraSparc.

So far, there has been no official statement from Microsoft about weaving support for x86-64 into their Windows operating systems. Although an Alpha-only version of Windows NT 4.0 was developed and released, the dismal monetary performance of this 64-bit chip has resulted in lower than forecasted earnings from the proprietary NT version. This experience would have definitely left the software giant with second thoughts, and it would closely evaluate whether or not attempting for yet another similar release will be worthwhile.

AMD's top brass assure the public that their chips will offer more promise to Microsoft's long term needs than the Itanium chip. There's substance behind these corporate assurances. Itanium is a niche product that can go only so far in a single focused market segment. But Hammer is a technology that encompasses the best of both worlds—64 and 32-bit computing. Even if Intel has plans to execute x86-64 architecture, without pushing it, the situation looks grim for the chip giant as its bothersome competitor is maintaining a one-step lead.

Hyperthreading

A technique used to make a single chip act like two, Hyperthreading is a concept introduced by Intel that lets a chip operate far more efficiently and almost as well as a dual processor machine by taking advantage of formerly unused circuitry on a Pentium 4 chip. With it, a computer can run a single application faster than a standard single processor system can.

Hyperthreading works by taking advantage of additional CPU registers (circuits that help manage data inside a chip) that exist on Pentium 4 processors, but aren't used. With the use of these registers, the chip is better able to manage its existing resources without the need to seek computing power elsewhere. Demos have shown a 30 per cent performance rise when this technique is actually used.

Software developers need to design their goods accordingly if they are to use Intel's power boosting methods, and that takes plenty of effort. So far, only 30 applications

have been optimised for the P4. Fortunately, Windows XP and Linux can both take advantage of the technology. Intel has requested the developer community to design applications, OSs and device drivers suitably.

According to Intel, instead of pushing clock speeds and tweaking parts of the chip for measurable performance gains, efficiency of operation is another factor that needs attention. Though this doesn't necessarily mean gaining the power of two chips, research into fields such as these could not only save money but also contribute to the long-term goal of better chip design down the road. Hyperthreading will make its presence felt in servers in 2002 and in desktops by 2003.

Linux has a patch that enables Hyperthreading on all Pentium 4 processors. Follow the link to download it (you will need to run gunzip for decompression)

<ftp://ftp.kernel.org/pub/linux/kernel/v2.4/testing/patch-2.4.17-pre6.gz>

Looking ahead

In computing, perhaps the most interesting times lie ahead. Progress in 64-bit processing will unlock many avenues and a lot more responsibility will be at stake with regards to who will lead the masses on this technology—the implications encompass a number of computing areas touching the lives of a lot of people.

At the moment, it seems like AMD will secure its place in history with its desktop ClawHammer chip. This will bring the long-awaited 64-bit technology to consumers. While it looks as though Intel has been keeping its mouth shut in response to x86-64, the chipmaker has a few trump cards up its sleeve, lest you forget the Itanium chip bombs.

The chip behemoth is looking at incorporating some capabilities in the next version of the Pentium family. Code-named Prescott, the chip is Intel's plan B that would extend 32-bit architecture to house 64-bit compatibility. Even if the cards are played right, Prescott could face an uphill battle. After spending massive amounts of time and money on constructing the Itanium, abruptly launching a chip with similar traits yet maintaining a higher degree of software compatibility would really undercut Intel's earlier efforts. A Pentium chip with 64-bit extensions could happen if AMD's ClawHammer gains credible market positioning. ■

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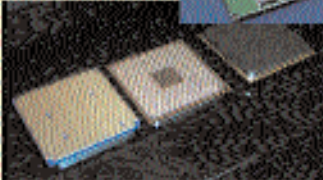
2001



Intel Itanium (64-bit)

IA-64, the underlying architecture of the Itanium processor is exclusive for 64 bit apps, leaving 32 bit apps out to dry. As a result its future is uncertain.

2002



AMD Hammer (64-bit)

Hammer is made up of ClawHammer and SledgeHammer, chips designed for the desktop and server respectively. Till date, these are the first 64-bit chips that are backward compatible, running both 64-bit and 32-bit applications.

Intel McKinley (64-bit)



With their 2nd generation 64-bit chip underway, Intel is shooting for the stars. The chip giant claims McKinley, due for release in mid 2002, will deliver up to twice the performance of Itanium.